

Mostafa Naseri

Machine Learning R&D Engineer | Efficient AI for Vision and Wireless Systems

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Professional Summary

Machine Learning R&D Engineer with a PhD in Electrical Engineering and experience taking models from research prototypes to resource-constrained deployments. Work spans fraud and anomaly detection, local-LLM agents, graph learning, optimized inference, learned image compression, and wireless signal processing, with international collaborations involving Princeton University, Nokia Bell Labs, Universidad de Málaga, and imec partners.

Technical Skills

Programming & Frameworks: Python, PyTorch, C/C++, scikit-learn, Keras, MATLAB

ML Methods & Architectures: Supervised and unsupervised anomaly detection, graph learning, gradient-boosted trees (LightGBM), transfer learning, contrastive learning, self-supervised learning; architectures — transformers, GNNs, CNNs, LSTMs

Agentic AI & LLMs: Local-LLM agents, agent workflow design, tool design and integration, task-specific fine-tuning

Inference & Production: TensorRT, torch.compile, ONNX, OpenVINO, llama.cpp; quantization, pruning, model compression, model packaging and production delivery, Kubernetes, Weights & Biases, Git

Vision & Wireless: Multimodal models, learned image compression, image tokenization, UWB AoA estimation, interference cancellation, 5G/O-RAN resource management

Professional & Research Experience

Machine Learning R&D Engineer, FraudBuster – Ghent, Belgium May 2025 – present

- Develop supervised, graph-based, and unsupervised anomaly-detection models for fraud and scam detection.
- Build local-LLM fraud-detection agents using custom tools and task-specific fine-tuning.
- Optimize and package models with their runtime dependencies as reproducible production bundles, balancing predictive performance, latency, memory use, and deployment constraints.

PhD Researcher in Electrical Engineering, Ghent University, IDLab-imec – Ghent, Belgium Apr 2021 – Apr 2025

- Developed efficient deep-learning systems for interference cancellation, UWB localization, and learned image transmission under channel and edge-compute constraints.
- Reduced MACs by **61.1%** while improving MSE by **0.63%** and reaching **800k symbols/s** in a fully convolutional interference-cancellation model.
- Built a Fast Fourier Convolution U-Net that improved MSE by **26.5%** with more than **76%** fewer operations and parameters than the benchmark.

Visiting Researcher, Princeton University – Princeton, NJ, USA July 2024 – Sept 2024

- Developed adaptive, progressive learned image-compression methods for robust low-latency wireless transmission in Prof. H. Vincent Poor's group.

Visiting Researcher, Universidad de Málaga, ERTIS Research Group – Málaga, Spain Apr 2024 – June 2024

- Investigated early-exit neural models for computationally efficient interference cancellation.

Selected Projects

MIT RF Challenge — ICASSP 2024 2024

Led an IDLab team recognized among the **top five teams worldwide** for data-driven radio-frequency signal separation.

- Built a time-frequency U-Net that improved mean-square-error performance by an average of **63%** over the baseline across four interference types.

EVOLVE — Progressive Learned Image Transmission 2024

Designed adaptive image-transmission pipelines using hyperprior and VQGAN compression models for dynamic wireless channels.

- Improved transmission latency by **38%** versus evaluated benchmarks; work later published in *npj Wireless Technology*.

5GECO — 5G Network Cross-Optimization 2022

Developed reinforcement-learning methods for multi-operator radio-resource management with Nokia Bell Labs, Citymesh, Accelleran, and University of Antwerp.

- Contributed to a deep Q-learning xApp that reduced SLA violations to **2.5%** for eMBB slices and eliminated them for URLLC slices in evaluation.

SmartWaterWay — Adaptive UWB Localization

2021

Developed transfer-learning and data-augmentation methods for environment-adaptive UWB angle-of-arrival estimation.

- Reduced estimation error by up to **96%** with approximately **10%** additional computation.

Education

Ghent University, PhD in Electrical Engineering – Ghent, Belgium

2021 – 2025

- Dissertation: *Towards Data-Driven Communication Signal Processing in Realistic Conditions*

Beijing University of Posts and Telecommunications, MSc in Electronics and Communication

2017 – 2019

Engineering – Beijing, China

- GPA: 90/100; BGS Scholarship for Outstanding Academic Performance

Shiraz University of Technology, BSc in Electrical Engineering – Shiraz, Iran

2010 – 2014

- GPA: 17.2/20

Selected Publications

Deep learning-based image compression for wireless communications: impacts on robustness, throughput, and latency

2026

Mostafa Naseri, Pooya Ashtari, Mohamed Seif, Eli De Poorter, H. Vincent Poor, Adnan Shahid

[10.1038/s44459-025-00019-6](https://doi.org/10.1038/s44459-025-00019-6) (npj Wireless Technology, 2, Article 14)

High-throughput adaptive co-channel interference cancellation for edge devices using depthwise separable convolutions, quantization, and pruning

2025

Mostafa Naseri, Eli De Poorter, Ingrid Moerman, H. Vincent Poor, Adnan Shahid

[10.1109/OJCOMS.2024.3523797](https://doi.org/10.1109/OJCOMS.2024.3523797) (IEEE Open Journal of the Communications Society, 6, 656–670)

Blind co-channel interference cancellation using fast Fourier convolutions

2024

Mostafa Naseri, Eli De Poorter, Ingrid Moerman, H. Vincent Poor, Adnan Shahid

[10.1109/VTC2024-SPRING62846.2024.10683131](https://doi.org/10.1109/VTC2024-SPRING62846.2024.10683131) (IEEE VTC-Spring)

A U-Net architecture for time-frequency interference signal separation of RF waveforms

2024

Mostafa Naseri, Jaron Fontaine, Ingrid Moerman, Eli De Poorter, Adnan Shahid

[10.1109/ICASSPW62465.2024.10626228](https://doi.org/10.1109/ICASSPW62465.2024.10626228) (2024 IEEE ICASSP Workshops, 91–92)

Machine learning-based angle of arrival estimation for ultra-wide band radios

2022

Mostafa Naseri, Adnan Shahid, Gert-Jan Gordebeke, Sam Lemey, Michiel Boes, Samuel Van de Velde, Eli De Poorter

[10.1109/LCOMM.2022.3167020](https://doi.org/10.1109/LCOMM.2022.3167020) (IEEE Communications Letters, 26(6), 1273–1277)

Selected Recognition

- Led a team selected among the top five worldwide — MIT RF Challenge, ICASSP 2024
- FI DGR Grant for Predoctoral Research Staff Recruitment — €23,774.97, Regional Government of Catalonia (2021)
- Ranked 119th among approximately 40,000 candidates — top 0.3% — in Iran's national MSc entrance examination (2014)
- BGS Scholarship for Academic Excellence — RMB 40,000 (approximately €5,200), BUPT (2018)

Earlier Telecom Experience

Nokia, Network Planning and Optimization Engineer (2016–2017): Monitored and optimized 3G/LTE sites and presented performance findings to MTN Irancell.

NAK Telecom Managed Services, Quality Assurance Engineer (2016): Analyzed 2G/3G/LTE network quality and presented customer improvement plans.